HW6

Greyson Knippel
MATH 339 — Gonzaga University
compiled with LaTeX

October 25, 2026

3.1 - 17

Let V be the set of all invertible 2×2 matrices, with the standard matrix addition and scalar multiplication.

claim: V is *not* a vector space, because it is not closed under addition.

counterexample: Take

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

both are invertible, since $\det A = 1 \neq 0$ and $\det B = 1 \neq 0$, so $A, B \in V$. but

$$A + B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \det(A + B) = 0,$$

so $A + B$ is not invertible and $A + B \notin V$.

therefore the closure axiom "if $\mathbf{u}, \mathbf{v} \in V$, then $\mathbf{u} + \mathbf{v} \in V$" doenst work, and V is not a vector space.

3.1 - 23

(a)

additive identity: we need $\mathbf{v}_s \oplus \mathbf{v}_t = \mathbf{v}_t$ for every t, i.e. $s + t = t$, so $s = 0$:

$$\mathbf{0} = \mathbf{v}_0 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

additive inverse: we need $\mathbf{v}_t \oplus \mathbf{v}_s = \mathbf{v}_0$, i.e. $t + s = 0$, so $s = -t$:

$$-\mathbf{v}_t = \mathbf{v}_{-t} = \begin{bmatrix} 1 - t \\ 2 + t \\ 3 - 2t \end{bmatrix}.$$

(b)

let $\mathbf{u} = \mathbf{v}_r$, $\mathbf{v} = \mathbf{v}_s$, $\mathbf{w} = \mathbf{v}_t$ be in V and $c, d \in \mathbb{R}$.

1. $\mathbf{u} \oplus \mathbf{v} = \mathbf{v}_{r+s} \in V$ since $r + s \in \mathbb{R}$.
2. $\mathbf{u} \oplus \mathbf{v} = \mathbf{v}_{r+s} = \mathbf{v}_{s+r} = \mathbf{v} \oplus \mathbf{u}$.
3. $(\mathbf{u} \oplus \mathbf{v}) \oplus \mathbf{w} = \mathbf{v}_{(r+s)+t} = \mathbf{v}_{r+(s+t)} = \mathbf{u} \oplus (\mathbf{v} \oplus \mathbf{w})$.
4. $\mathbf{v}_0 \oplus \mathbf{u} = \mathbf{v}_{0+r} = \mathbf{v}_r = \mathbf{u}$, from (a).

5. $\mathbf{u} \oplus \mathbf{v}_{-r} = \mathbf{v}_{r-r} = \mathbf{v}_0$, from (a).
6. $c \odot \mathbf{u} = \mathbf{v}_{cr} \in V$ since $cr \in \mathbb{R}$.
7. $c \odot (\mathbf{u} \oplus \mathbf{v}) = \mathbf{v}_{c(r+s)} = \mathbf{v}_{cr+cs} = (c \odot \mathbf{u}) \oplus (c \odot \mathbf{v})$.
8. $(c+d) \odot \mathbf{u} = \mathbf{v}_{(c+d)r} = \mathbf{v}_{cr+dr} = (c \odot \mathbf{u}) \oplus (d \odot \mathbf{u})$.
9. $c \odot (d \odot \mathbf{u}) = \mathbf{v}_{c(dr)} = \mathbf{v}_{(cd)r} = (cd) \odot \mathbf{u}$.
10. $1 \odot \mathbf{u} = \mathbf{v}_{1 \cdot r} = \mathbf{v}_r = \mathbf{u}$.

all 10 of the checks hold, so V is a vector space.

(c)

for any $\mathbf{v} = \mathbf{v}_t \in V$,

$$0 \odot \mathbf{v}_t = \begin{bmatrix} 1+0t \\ 2-0t \\ 3+2(0)t \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \mathbf{v}_0 = \mathbf{0}.$$

3.2 - 9

S is a subspace of $\mathbb{R}^3$ if it's got $\mathbf{0}$ and is closed under addition and scalar multiplication.

zero vector: take $s = t = 0$:

$$\begin{bmatrix} 0-2(0) \\ 0 \\ 0+0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \in S.$$

closed under addition: let

$$\mathbf{u} = \begin{bmatrix} s_1 - 2t_1 \\ s_1 \\ t_1 + s_1 \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} s_2 - 2t_2 \\ s_2 \\ t_2 + s_2 \end{bmatrix}.$$

then

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} (s_1 + s_2) - 2(t_1 + t_2) \\ s_1 + s_2 \\ (t_1 + t_2) + (s_1 + s_2) \end{bmatrix},$$

which has the form of S with $s = s_1 + s_2$ and $t = t_1 + t_2$, so $\mathbf{u} + \mathbf{v} \in S$.

closed under scalar multiplication: for $c \in \mathbb{R}$,

$$c\,\mathbf{u} = \begin{bmatrix} cs_1 - 2(ct_1) \\ cs_1 \\ ct_1 + cs_1 \end{bmatrix},$$

which has the form of S with $s = cs_1$ and $t = ct_1$, so $c\,\mathbf{u} \in S$.

all three of the checks hold, so S is a subspace of $\mathbb{R}^3$.

3.2 - 41

$$H = \left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \right\}$$

$$T = \left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} -3 \\ 4 \\ 5 \end{bmatrix} \right\}$$

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 6 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ 4 \\ 5 \end{bmatrix} \right\}$$

key fact: put the vectors of H in the columns of

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 3 & 2 \\ 2 & -1 & -1 \end{bmatrix}.$$

then

$$\det A = 1 \begin{vmatrix} 3 & 2 \\ -1 & -1 \end{vmatrix} - (-1) \begin{vmatrix} 2 & 2 \\ 2 & -1 \end{vmatrix} + 1 \begin{vmatrix} 2 & 3 \\ 2 & -1 \end{vmatrix} = (-1) + (-6) + (-8) = -15 \neq 0.$$

so A is invertible, which means $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b}$ (so $\text{span}(H) = \mathbb{R}^3$) and $A\mathbf{x} = \mathbf{0}$ has only $\mathbf{x} = \mathbf{0}$ (so H is linearly independent).

(a)

$H \subseteq S$, so $\mathbb{R}^3 = \text{span}(H) \subseteq \text{span}(S) \subseteq \mathbb{R}^3$. therefore $\text{span}(S) = \mathbb{R}^3$.
so basically everything in S can be built in H.

(b)

no. S has 5 vectors in $\mathbb{R}^3$

(c)

$\text{span}(T) = \mathbb{R}^3$, since $H \subseteq T$ (same reasoning as (a)). T is not linearly independent

(d)

$\text{span}(H) = \mathbb{R}^3$ and H is linearly independent

3.3 - 19

split the vector into an s part and a t part:

$$\begin{bmatrix} s + 2t \\ -s + t \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}.$$

so $S = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \right\}$.

linearly independent: suppose

$$a \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

the third entry gives $b = 0$, then the first entry gives $a = 0$.
so a basis for S is

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \right\}, \qquad \dim S = 2.$$

3.3 - 35

$\dim \mathbb{R}^3 = 3$, so we need one more vector. the two vectors that are given to us are already linearly independent, since neither is a multiple of the other. we can add $\mathbf{e}_1$:

$$A = \begin{bmatrix} -1 & 1 & 1 \\ 1 & 1 & 0 \\ 3 & 1 & 0 \end{bmatrix}, \qquad \det A = 1 \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} = 1 - 3 = -2 \neq 0$$

(expanding along the third column). so the three columns are linearly independent, and the basis is:

$$\left\{ \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}.$$